

- Fragmentation of packets (*-f* and *-ff* options)
- Using the 'FTP bounce' technique to scan via FTP servers (*-b*)
- Changing the scan delay (*-scan_delay* is especially useful if the target has IDS/IPS, and blocks scan requests)
- Timing policies (*-T*)
- Scripted scans

Listing active hosts on the network

A common sequence in network testing is to first generate a list of all active hosts in a network. The list can be used as an input to other applications, which lack the capacity to scan for active hosts but must be given one or more target IP address. Here is a short command sequence that does this:

```
nmmap -sP -n -oG hostlist 192.168.10.0/24
cut -d ' ' -f2 hostlist > iplist
```

The first command executes a ping scan (*-sP*) and generates a list of active hosts in the target range/network. This list will be stored in the file *hostlist* in *greppable* format (*-oG*). The second command reads the data from the file, splits each line into fields based on a space character delimiter, and outputs the second field (the IP address), thus generating a new file, *iplist*, which is simply the list of active IP addresses/hosts in the given range.

The Zenmap GUI

Nmap has a huge list of command line options that are difficult to remember and use, even for experienced administrators. Zenmap, a GUI for Nmap, simplifies this considerably. It even provides ready-made scan profiles from which you can choose. The commands you generate in the GUI can also be run at a command line, if required. The GUI also has a very important added function—a graphical display of discovered hosts, and the hops required to reach each host! A sample output from this feature is shown in Figure 1.

We hope that the few important concepts about the TCP/IP protocol, the power of Nmap, and the other ideas interested you! Next month, we'll look at another interesting networking tool, so watch this column! **END** 

References

- www.nmap.org/book
- www.nmap.org
- Nmap in the Enterprise: Your Guide to Network Scanning by Angela Orebrough and Becky Pinkard

By: Rajesh Deodhar

The author is a Bachelor of Engineering in industrial electronics (BE), certified information systems auditor (CISA) and Diploma in Cyber Law (DCL). He has more than 15 years of experience in the field of computer hardware, networking and firewalls. He is a director at Omega Systems and Services Pvt Ltd, Pune, and can be contacted at rajesh@omegasystems.co.in

LINUX

ForYou

Calling...

Share your knowledge

Inviting FOSS experts to write articles on their area of interest

LFFY covers a myriad of topics—network management, software development, embedded systems, community issues, and even hands-on guide for newbies. If you've got an interesting topic, let us know. Thanks to the launch of www.openTis.com (aka linuxforu.com) we are now trying to extend our content portfolio related to Linux & Open Source.

To know more on how to become an LFFY author, contact us at lfyedit@efyindia.com

Few topics that top our list:

- ♦ Tips 'n' Tricks for software developers or IT implementers
- ♦ Cool tweaks for FOSS enthusiasts
- ♦ FOSS on mobile
- ♦ Virtualisation (Implementation)
- ♦ OpenJDK or Java on Linux
- ♦ OpenSolaris (software development)
- ♦ How can I do 'that' on Linux
- ♦ Reviews of latest open source projects & tools

